ZEBRAFISH KNOWLEDGE SERIES

Why Zebrafish Conservation Matters

Wild populations, freshwater habitats and the genetic variation that research depends on

The zebrafish is common and not threatened, yet the rivers it lives in are among the most degraded habitats on Earth, and the wild populations hold genetic variation that laboratory strains lack.

1 The question

The zebrafish is not an endangered species. The reasons to conserve it lie in its habitat and in its genetic variation.

The IUCN Red List classifies *Danio rerio* as Least Concern. It is common across much of South Asia, lives in a wide range of conditions, and tolerates water temperatures from about 6 to 38 °C. Surveys have recorded it at elevations up to 1,795 m. It also survives well in farmland, including ditches and rice paddies.

Conserving the zebrafish therefore does not mean rescuing a vanishing species. It means keeping the freshwater systems it depends on in good condition, and keeping the natural variation that the species still holds.

2 The state of freshwater life

The Living Planet Index, produced by the Zoological Society of London and published with WWF's Living Planet Report 2024, tracks 34,836 population trends across 5,495 species. Between 1970 and 2020 monitored wildlife populations fell by an average of 73 per cent. Freshwater populations, covering fish, birds, mammals, amphibians and reptiles, fell by 85 per cent, the steepest decline of any environment.

Figure 1. Average decline in monitored vertebrate populations. Source: Living Planet Index 2024, Zoological Society of London.

The main threats listed in the report are habitat loss and degradation, overexploitation, invasive species, pollution, climate change and disease. Freshwater habitats are exposed to all of them at once, because rivers collect the effects of everything that happens upstream.

3 A genetic reserve for science

The zebrafish used in laboratories descend from a small number of founders and have been bred for many generations. In 2011 Whiteley and colleagues compared 13 wild populations from Nepal, India and Bangladesh with three widely used laboratory strains, using 1,832 genetic markers and a mitochondrial gene.

FINDING	RESULT
Mitochondrial lineages in the wild	Three major clades that differ by about 5.8% on average
Nuclear markers in the wild	Five genetically distinct groups, matching separate geographic regions (overall $F_{ST} = 0.170$)
Laboratory strains	Two further groups, distinct from all wild populations
Signals of selection	71 loci (3.4%) strongly divergent, some probably reflecting selection in the laboratory

> **From the study**
>
> *"Substantial genetic variation that exists in zebrafish as a whole is missing from lab strains that we analysed."*
>
> Whiteley et al., *Molecular Ecology*, 2011.

This gives research a practical reason to protect wild populations. The lab strains are a small sample of what the species holds. Wild fish carry variants that may affect disease resistance, development and tolerance of stress, and they can only be studied while they persist in the places where they evolved.

4 A sign of water quality

Wild zebrafish are typically annual and breed in shallow, slow water at the edges of streams and floodplains. A short life cycle and a close link to such habitat mean their numbers can change quickly when the water is polluted, drained or silted up. Fish surveys that include zebrafish and related small species are a practical way for local organisations to monitor the health of a stream.

5 Pressures on the habitat

A survey of the Narayani River system in Chitwan, Nepal, by the Fisheries Development Centre, Hetauda, listed the pressures facing its fish. They include unregulated fishing, dynamiting and poisoning, fishing of broodstock during spawning, nets with a mesh below 2 cm, and the collection of fish fry. They also include deforestation leading to erosion and silting, and pollution from industrial waste and agricultural chemicals such as insecticides and pesticides.

These pressures are common to rivers across the region. Because zebrafish live in the same shallow, warm margins as farmland and villages, they meet them early.

6 What conservation looks like in practice

- **Survey and record.** Document where wild zebrafish occur, at what elevation and season, and in what water conditions.
- **Protect shallow margins.** Keep stream edges, ponds and floodplain wetlands connected and free of drainage and dredging.
- **Reduce chemical inputs.** Promote integrated pest management and sensible buffer zones beside watercourses.
- **Stop destructive fishing.** Support enforcement against poisoning and dynamiting, and encourage community fishing rules.
- **Conserve genetic diversity.** Collect tissue samples from wild populations for genetic study and, where appropriate, build wild-derived research stocks.
- **Work with communities.** Rivers and paddies are used by farmers and fishers, so protection works best when it also improves their livelihoods.

Sources

IUCN Red List status (Least Concern), range and elevation: Wikipedia, Zebrafish; Animal Diversity Web, *Danio rerio*.

Zoological Society of London (2024). Living Planet Index 2024 (zsl.org/news-and-events/news/living-planet-index-2024); WWF Living Planet Report 2024.

Whiteley A.R. et al. (2011). Population genomics of wild and laboratory zebrafish (*Danio rerio*). *Molecular Ecology* 20(20): 4259–4276.

Dhital R.R., Fisheries Development Centre, Hetauda, Nepal. Fish survey of the Narayani River system, Chitwan (FAO, fao.org/4/y3994e/y3994e0e.htm).